Task 7. Data compressing algorithms.

Goal

The use of data compressing algorithms on textual data.

Problems and methods

1. Take your favorite chapter from "The Song of Hiawatha" by Henry Wadsworth Longfellow https://www.hwlongfellow.org/poems_poem.php?pid=62 .

2. Use the Huffman algorithm to compress the ascii encoded text to a binary sequence.

3. Calculate the compression ratio (ascii has 8 bits per symbol) and the size of your code table.

4. Do the same with Lempel-Ziv-Welch algorithm.

5. Analyze the results and compare them.

6. Try to invent and explain an algorithm for embedding the name of a state (from Afghanistan to Zimbabwe https://en.wikipedia.org/wiki/List_of_sovereign_states) in some world-wide data-set for some ML task. No code required. Only idea and its brief description.